

How does national well-being Shape Global Music Consumption in 2024?

The 3 datasets used:

- Global streaming data from spotify for 2024
- World happiness report 2024
- World happiness report 2023

Hypothesis

H1: Populations facing higher economic uncertainty will exhibit higher levels of repetitive music consumption.

Rationale: Research shows that individuals often prefer familiar music over new music (Pereira et al., 2011). This implies in times of stress and unpredictability, individuals will choose more familiar music rather than experiment. This was quantified through assessing how many times songs were repeatedly streamed.

Independent Variable: Economic Uncertainty Index

Dependent Variable: Spotify Repetition Score (Streams per Listener)

H2: Populations with lower levels of social support and higher economic uncertainty will consume a higher proportion of positive music.

Rationale: This hypothesis was based on research conducted by de Lucio & Palomeque (2024) who found that when there is economic uncertainty and low social support, individuals lack external resources for mood regulation and rely on positive music instead. There exists a relatively novel term for this phenomenon known as recession pop which suggests that the rise of consumption in upbeat or pop music can be a recession indicator (Dickler & Solá, 2024).

Independent Variable: Economic Uncertainty Index and Social Support Index

Dependent Variable: Positive Music Share

H3: Higher economic uncertainty correlates with a preference for older music.

Rationale: Uncertainty drives a psychological retreat into the safe past. Listeners in uncertain economies will stream music with older average release years to evoke simpler, more stable times. Gibbs & Egermann (2021) found that nostalgic music boosts positive emotions and can be used as an emotion regulation technique.

Independent Variable: Economic Uncertainty Index

Dependent Variable: Nostalgia Index

Motivation

To test the three hypothesis established by research above where populations facing economic uncertainty or low social support regulate their mood through specific media consumption habits for the year 2024.

Data Cleaning

1. The first step was importing the datasets in R. The sheets were all converted to csvs for ease.

```
# | echo: false
library(tidyverse)
library(ggplot2)
library(ggthemes)
library(ggrepel)
library(corrplot)
spotify_dataset <- read.csv("Spotify_2024_Global_Streaming_Data.csv")
happiness_report_2024 <- read.csv("WHR24_Data_Figure_2.1.csv")
happiness_report_2023 <- read.csv("WHR23_Data_Figure_2.1.csv")
```

2. Then only relevant columns which were of interest from both reports were selected. These included the total score and the components explained by different factors. Then the columns were renamed to make the datasets easier to work with.

```
happiness_report_2023 <- happiness_report_2023 %>%
  select(Country.name, Ladder.score, Explained.by..Log.GDP.per.capita,
         Explained.by..Freedom.to.make.life.choices,
         Explained.by..Generosity, Explained.by..Social.support,
         Explained.by..Healthy.life.expectancy,
         Explained.by..Perceptions.of.corruption, Dystopia...residual) %>%
```

```

rename(GDP_per_capita = Explained.by..Log.GDP.per.capita,
       Freedom_life_choices = Explained.by..Freedom.to.make.life.choices,
       Generosity = Explained.by..Generosity,
       Social_support = Explained.by..Social.support,
       Healthy_life_expectancy = Explained.by..Healthy.life.expectancy,
       Corruption = Explained.by..Perceptions.of.corruption,
       Dystopia = Dystopia...residual)

happiness_report_2024 <- happiness_report_2024 %>%
  select(!c(upperwhisker, lowerwhisker)) %>%
  rename(GDP_per_capita = Explained.by..Log.GDP.per.capita,
       Freedom_life_choices = Explained.by..Freedom.to.make.life.choices,
       Generosity = Explained.by..Generosity,
       Social_support = Explained.by..Social.support,
       Healthy_life_expectancy = Explained.by..Healthy.life.expectancy,
       Corruption = Explained.by..Perceptions.of.corruption,
       Dystopia = Dystopia...residual)

spotify_dataset <- spotify_dataset %>%
  rename(Release_Year = Release.Year,
       Monthly_Listeners_Millions = Monthly.Listeners..Millions.,
       Total_Streams_Millions = Total.Streams..Millions.,
       Platform_Type = Platform.Type,
       Skip_Rate = Skip.Rate....,
       Total_hours_Streamed = Total.Hours.Streamed..Millions.)

```

3. Then the countries of interest were identified from the spotify dataset using unique since the dataset includes several entries for each country and these names were saved as a vector.

The countries were sorted alphabetically in the happiness report datasets to find the countries of interest and check which ones differed from the spotify dataset. Then case when was used to fix the discrepancies in the names of countries across the datasets and the relevant countries were filtered out from the happiness datasets.

```

countries_of_interest <- unique(spotify_dataset$Country)
countries_of_interest

```

```

[1] "Germany"      "Brazil"       "United States" "Italy"
[5] "Sweden"       "France"       "United Kingdom" "Argentina"
[9] "South Korea"  "South Africa" "Australia"      "Netherlands"
[13] "India"        "Canada"       "Mexico"         "Spain"

```

[17] "Japan" "Indonesia" "Russia" "Turkey"

```
happiness_report_2023 <- happiness_report_2023 %>%
  arrange(Country.name) %>%
  mutate(Country.name = case_when(
    Country.name == "Turkiye" ~ "Turkey",
    TRUE ~ Country.name)) %>%
  filter(Country.name %in% countries_of_interest)

happiness_report_2023
```

	Country.name	Ladder.score	GDP_per_capita	Freedom_life_choices	Generosity
1	Argentina	6.024	1.590	0.587	0.088
2	Australia	7.095	1.899	0.677	0.242
3	Brazil	6.125	1.454	0.558	0.131
4	Canada	6.961	1.881	0.656	0.218
5	France	6.661	1.856	0.582	0.083
6	Germany	6.892	1.919	0.618	0.153
7	India	4.036	1.159	0.685	0.175
8	Indonesia	5.277	1.384	0.663	0.422
9	Italy	6.405	1.832	0.438	0.097
10	Japan	6.129	1.825	0.556	0.009
11	Mexico	6.330	1.550	0.632	0.086
12	Netherlands	7.403	1.942	0.672	0.251
13	Russia	5.661	1.680	0.449	0.120
14	South Africa	5.275	1.417	0.464	0.090
15	South Korea	5.951	1.853	0.446	0.112
16	Spain	6.436	1.798	0.533	0.101
17	Sweden	7.395	1.921	0.754	0.225
18	Turkey	4.614	1.714	0.125	0.095
19	United Kingdom	6.796	1.857	0.626	0.272
20	United States	6.894	1.980	0.557	0.210

	Social_support	Healthy_life_expectancy	Corruption	Dystopia
1	1.388	0.427	0.082	1.861
2	1.497	0.532	0.310	1.938
3	1.250	0.387	0.137	2.205
4	1.484	0.541	0.364	1.815
5	1.433	0.566	0.270	1.872
6	1.401	0.539	0.365	1.898
7	0.674	0.252	0.111	0.979
8	1.169	0.314	0.038	1.288
9	1.365	0.559	0.063	2.052

10	1.396	0.622	0.207	1.513
11	1.169	0.389	0.115	2.389
12	1.488	0.545	0.394	2.110
13	1.383	0.366	0.091	1.572
14	1.428	0.149	0.019	1.708
15	1.188	0.603	0.163	1.587
16	1.491	0.567	0.157	1.789
17	1.510	0.562	0.520	1.903
18	1.148	0.467	0.096	0.969
19	1.366	0.511	0.340	1.822
20	1.460	0.390	0.172	2.124

```
happiness_report_2024 <- happiness_report_2024 %>%
  arrange(Country.name) %>%
  mutate(Country.name = case_when(
    Country.name == "Turkiye" ~ "Turkey",
    TRUE ~ Country.name)) %>%
  filter(Country.name %in% countries_of_interest)

happiness_report_2024
```

	Country.name	Ladder.score	GDP_per_capita	Social_support
1	Argentina	6.188	1.562	1.381
2	Australia	7.057	1.854	1.461
3	Brazil	6.272	1.430	1.269
4	Canada	6.900	1.840	1.459
5	France	6.609	1.818	1.348
6	Germany	6.719	1.871	1.390
7	India	4.054	1.166	0.653
8	Indonesia	5.568	1.361	1.184
9	Italy	6.324	1.800	1.328
10	Japan	6.060	1.786	1.354
11	Mexico	6.678	1.521	1.241
12	Netherlands	7.319	1.901	1.462
13	Russia	5.785	1.642	1.351
14	South Africa	5.422	1.389	1.369
15	South Korea	6.058	1.815	1.178
16	Spain	6.421	1.766	1.471
17	Sweden	7.344	1.878	1.501
18	Turkey	4.975	1.702	1.175
19	United Kingdom	6.749	1.822	1.326
20	United States	6.725	1.939	1.392

	Healthy_life_expectancy	Freedom_life_choices	Generosity	Corruption	Dystopia
1	0.585	0.681	0.087	0.080	1.812
2	0.692	0.756	0.225	0.323	1.745
3	0.548	0.685	0.130	0.142	2.068
4	0.701	0.730	0.230	0.368	1.572
5	0.727	0.650	0.112	0.281	1.672
6	0.702	0.700	0.174	0.368	1.513
7	0.417	0.767	0.174	0.122	0.756
8	0.472	0.779	0.399	0.055	1.318
9	0.720	0.513	0.112	0.074	1.777
10	0.785	0.632	0.023	0.219	1.261
11	0.544	0.722	0.086	0.127	2.437
12	0.706	0.725	0.247	0.372	1.906
13	0.531	0.551	0.138	0.121	1.451
14	0.322	0.537	0.078	0.034	1.693
15	0.770	0.555	0.126	0.158	1.456
16	0.729	0.619	0.119	0.177	1.540
17	0.724	0.838	0.221	0.524	1.658
18	0.631	0.202	0.068	0.115	1.083
19	0.672	0.713	0.267	0.351	1.598
20	0.542	0.586	0.223	0.169	1.873

Data Wrangling

1. The two happiness report datasets from 2023 and 2024 were joined using a full join to ensure no country was dropped in case of minor mismatches across reports. The suffix was changed from x and y to make it easier to handle later and clearer to understand.

```
joined_whr <- full_join(x = happiness_report_2023,
                        y = happiness_report_2024,
                        by = "Country.name",
                        suffix = c(".2023", ".2024"))
```

2. The genres in the spotify dataset were converted to factor. Then using the unique function, all the genres were assessed. They were then classified into two categories upbeat music and mellow music. These categories were saved as individual values. Then a new column was added called category and the respective categories were added using case when. Then the columns that were not needed removed.

```
spotify_dataset$Genre <- as.factor(spotify_dataset$Genre)

unique(spotify_dataset$Genre)
```

```
[1] K-pop      R&B      Reggaeton Indie      Rock      Pop      Jazz
[8] Hip Hop    EDM      Classical
Levels: Classical EDM Hip Hop Indie Jazz K-pop Pop R&B Reggaeton Rock
```

```
Upbeat_genres <- c("Pop", "K-pop", "EDM", "Reggaeton", "Hip Hop")
Mellow_genres <- c("Rock", "R&B", "Jazz", "Classical", "Indie")
```

```
spotify_dataset_modified <- spotify_dataset %>%
  mutate(Category = case_when(
    Genre %in% Upbeat_genres ~ "Upbeat",
    Genre %in% Mellow_genres ~ "Mellow",
    TRUE ~ "Other"
  )) %>%
  select(!c(Artist, Album, Avg.Stream.Duration..Min.))
```

3. The dependent variables were constructed from the spotify dataset by creating 4 scores. The data was grouped by country. The first the repetition score to test comfort listening. This was calculated by dividing the sum of the total streams by the sum of monthly listeners. Then the nostalgia index was calculated by taking difference between the weighted mean if the release year in respect to the total hours streamed and 2024. The avg skip rate was calculated by the weighted mean of the skip rate against the total streams. Finally, the positive share was calculated as the proportion of total listening hours devoted to upbeat music.

```
metrics <- spotify_dataset_modified %>%
  group_by(Country) %>%
  summarise(
    Repetition_Score = sum(Total_Streams_Millions) /
    sum(Monthly_Listeners_Millions),
    nostalgia_index = 2024 - weighted.mean(Release_Year, w = Total_hours_Streamed),
    Avg_Skip_Rate = weighted.mean(Skip_Rate, w = Total_Streams_Millions),
    Positive_Hours = sum(Total_hours_Streamed[Category == "Upbeat"])/
    sum(Total_hours_Streamed),
    .groups = "drop")
```

4. Next the changes from the year 2023 to 2024 were calculating by subtracting each column in the joined dataset. This was done to quantify the increase or decrease in the national well-being.

```
whr_changes <- joined_whr %>%
  mutate(delta_ladder = Ladder.score.2024 - Ladder.score.2023,
    delta_gdp = GDP_per_capita.2024 - GDP_per_capita.2023,
```

```

delta_freedom = Freedom_life_choices.2024 - Freedom_life_choices.2023,
delta_generosity = Generosity.2024 - Generosity.2023,
delta_social_support = Social_support.2024 - Social_support.2023,
delta_hle = Healthy_life_expectancy.2024 - Healthy_life_expectancy.2023,
delta_corruption = Corruption.2024 - Corruption.2023,
delta_dystopia = Dystopia.2024 - Dystopia.2023) %>%
select(Country.name, delta_ladder,
       delta_gdp,delta_corruption, delta_dystopia,
       delta_generosity, delta_social_support,
       delta_freedom, delta_hle)
whr_changes

```

	Country.name	delta_ladder	delta_gdp	delta_corruption	delta_dystopia
1	Argentina	0.164	-0.028	-0.002	-0.049
2	Australia	-0.038	-0.045	0.013	-0.193
3	Brazil	0.147	-0.024	0.005	-0.137
4	Canada	-0.061	-0.041	0.004	-0.243
5	France	-0.052	-0.038	0.011	-0.200
6	Germany	-0.173	-0.048	0.003	-0.385
7	India	0.018	0.007	0.011	-0.223
8	Indonesia	0.291	-0.023	0.017	0.030
9	Italy	-0.081	-0.032	0.011	-0.275
10	Japan	-0.069	-0.039	0.012	-0.252
11	Mexico	0.348	-0.029	0.012	0.048
12	Netherlands	-0.084	-0.041	-0.022	-0.204
13	Russia	0.124	-0.038	0.030	-0.121
14	South Africa	0.147	-0.028	0.015	-0.015
15	South Korea	0.107	-0.038	-0.005	-0.131
16	Spain	-0.015	-0.032	0.020	-0.249
17	Sweden	-0.051	-0.043	0.004	-0.245
18	Turkey	0.361	-0.012	0.019	0.114
19	United Kingdom	-0.047	-0.035	0.011	-0.224
20	United States	-0.169	-0.041	-0.003	-0.251
	delta_generosity	delta_social_support	delta_freedom	delta_hle	
1	-0.001	-0.007	0.094	0.158	
2	-0.017	-0.036	0.079	0.160	
3	-0.001	0.019	0.127	0.161	
4	0.012	-0.025	0.074	0.160	
5	0.029	-0.085	0.068	0.161	
6	0.021	-0.011	0.082	0.163	
7	-0.001	-0.021	0.082	0.165	

8	-0.023	0.015	0.116	0.158
9	0.015	-0.037	0.075	0.161
10	0.014	-0.042	0.076	0.163
11	0.000	0.072	0.090	0.155
12	-0.004	-0.026	0.053	0.161
13	0.018	-0.032	0.102	0.165
14	-0.012	-0.059	0.073	0.173
15	0.014	-0.010	0.109	0.167
16	0.018	-0.020	0.086	0.162
17	-0.004	-0.009	0.084	0.162
18	-0.027	0.027	0.077	0.164
19	-0.005	-0.040	0.087	0.161
20	0.013	-0.068	0.029	0.152

5. A z function was created that scaled the values to create the indices. The differences were scaled using the z function. For worsened outcomes the scaling was inversed to better construct the economic uncertainty index. Then the indices were created by first binding the columns for the parameters of interest and then finding their row means. Then the country names and the calculated indices were selected for further analysis.

```
z <- function(x) {
  as.numeric(scale(x))
}

whr_indices <- whr_changes %>%
  mutate(z_gdp = z(x = delta_gdp),
         z_gdp_worsen = - z(x = delta_gdp),
         z_corruption_worsen = - z(x = delta_corruption),
         z_dystopia_worsen = - z(x = delta_dystopia),
         z_social_support = z(x = delta_social_support),
         z_generosity = z(x = delta_generosity),
         z_hle = z(x = delta_hle),
         z_freedom = z(x = delta_freedom),
         economics_uncertainty_index = rowMeans(
           cbind(z_gdp_worsen, z_corruption_worsen, z_dystopia_worsen)),
         social_support_index = rowMeans(
           cbind(z_social_support, z_generosity)),
         life_eval_index = rowMeans(
           cbind(z_hle, z_freedom))) %>%
  select(Country.name, economics_uncertainty_index,
         social_support_index, life_eval_index)

whr_indices
```

	Country.name	economics_uncertainty_index	social_support_index
1	Argentina	-0.09751414	0.05174364
2	Australia	0.27545000	-0.88341803
3	Brazil	-0.18233349	0.42081115
4	Canada	0.57282446	0.22158678
5	France	0.17018861	-0.07387869
6	Germany	1.15982880	0.71478950
7	India	-0.95160330	-0.14698502
8	Indonesia	-1.00875498	-0.35579395
9	Italy	0.20955815	0.14940595
10	Japan	0.30291890	0.04571209
11	Mexico	-0.74778025	1.20586041
12	Netherlands	1.25364509	-0.31611757
13	Russia	-0.60969620	0.31853849
14	South Africa	-0.69895020	-1.04630413
15	South Korea	0.47095709	0.49994903
16	Spain	-0.12986216	0.48887734
17	Sweden	0.63062011	-0.07480419
18	Turkey	-1.57862531	-0.31633246
19	United Kingdom	0.15441396	-0.54756557
20	United States	0.80471486	-0.35607476
	life_eval_index		
1		-0.158757633	
2		-0.280852229	
3		0.958923928	
4		-0.398043532	
5		-0.423933439	
6		0.133681518	
7		0.363160828	
8		0.356884098	
9		-0.259865616	
10		-0.006948045	
11		-0.596729640	
12		-0.775507346	
13		0.831926037	
14		1.070133725	
15		1.225473171	
16		0.112694904	
17		0.065818383	
18		0.131229871	
19		0.021393510	
20		-2.370682494	

6. Then both the modified datasets were joined by the country names.

```
joined_genre_well_being <- full_join(x = metrics,  
                                     y = whr_indices,  
                                     by = c("Country" = "Country.name"))  
  
joined_genre_well_being
```

A tibble: 20 x 8

	Country <chr>	Repetition_Score <dbl>	nostalgia_index <dbl>	Avg_Skip_Rate <dbl>	Positive_Hours <dbl>
1	Argentina	44.4	3.90	22.7	0.585
2	Australia	48.4	3.64	18.6	0.577
3	Brazil	49.4	3.86	20.2	0.464
4	Canada	42.4	3.02	20.5	0.494
5	France	70.5	3.43	18.4	0.477
6	Germany	77.9	2.89	14.6	0.591
7	India	48.0	3.83	18.6	0.515
8	Indonesia	37.7	3.05	24.2	0.393
9	Italy	55.7	3.42	22.1	0.468
10	Japan	36.6	2.43	22.7	0.553
11	Mexico	53.2	3.94	19.1	0.394
12	Netherlands	46.4	3.46	13.6	0.356
13	Russia	43.6	3.95	21.5	0.349
14	South Africa	56.2	3.70	22.5	0.571
15	South Korea	66.0	3.37	20.0	0.438
16	Spain	40.3	3.37	17.6	0.363
17	Sweden	61.4	3.34	23.8	0.503
18	Turkey	53.7	3.94	20.1	0.266
19	United Kingdom	44.8	3.65	19.3	0.579
20	United States	63.6	2.87	18.4	0.450

i 3 more variables: economics_uncertainty_index <dbl>,
social_support_index <dbl>, life_eval_index <dbl>

Data Visualization

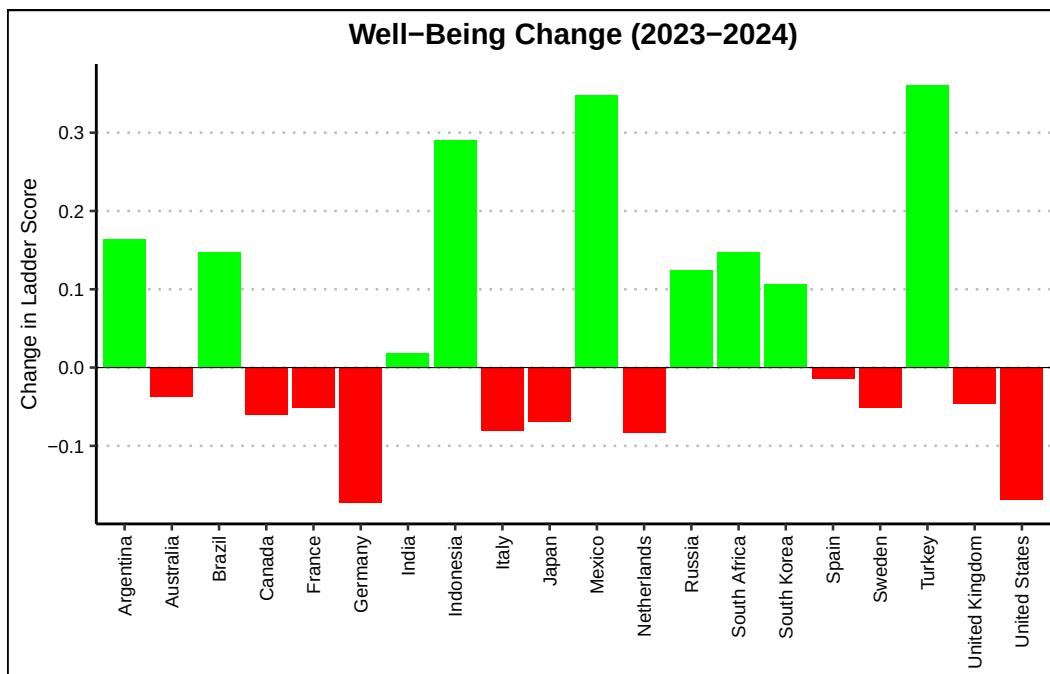
1. Well-being change plot

```
ggplot(data = whr_changes,  
       mapping = aes(x = Country.name,  
                     y = delta_ladder,
```

```

    fill = delta_ladder > 0)) +
  geom_col(show.legend = FALSE) +
  geom_hline(yintercept = 0, color = "black", size = 0.2) +
  scale_fill_manual(values = c("TRUE" = "green", "FALSE" = "red")) +
  labs(
    title = "Well-Being Change (2023-2024)",
    x = NULL,
    y = "Change in Ladder Score") +
  theme_clean(base_size = 10) +
  theme(axis.text.x = element_text(angle = 90,
                                     hjust = 1,
                                     vjust = 0.5),
        plot.title = element_text(hjust = 0.5))

```



Out of the 20 countries, 9 showed an increase in overall well-being from 2023 to 2024 while 11 showed a decrease. To understand the impact on each component this plot was further faceted by individual metrics.

2. Well-being change plot for each component

```

whr_plot_data <- whr_changes %>%
  select(!delta_ladder) %>%
  pivot_longer(

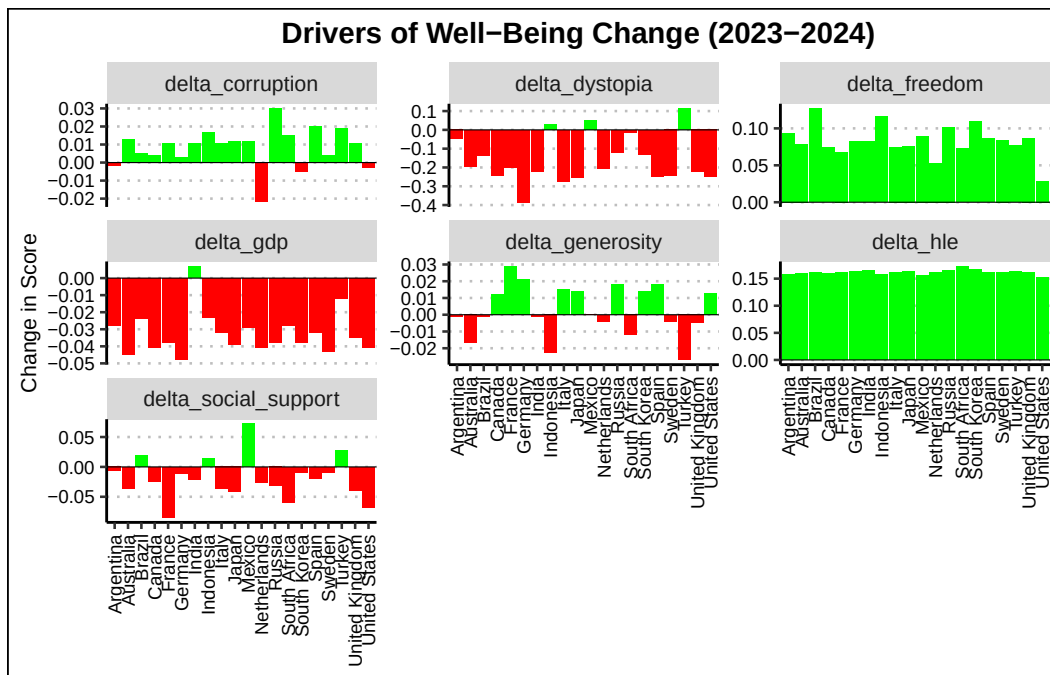
```

```

cols = starts_with("delta"),
names_to = "Metric",
values_to = "Change")

ggplot(data = whr_plot_data,
       mapping = aes(x = Country.name,
                     y = Change,
                     fill = Change > 0)) +
  geom_col(show.legend = FALSE) +
  geom_hline(yintercept = 0, color = "black", size = 0.2) +
  facet_wrap(~Metric, scales = "free_y") +
  scale_fill_manual(values = c("TRUE" = "green", "FALSE" = "red")) +
  labs(title = "Drivers of Well-Being Change (2023-2024)",
       x = NULL,
       y = "Change in Score") +
  theme_clean(base_size = 10) +
  theme(axis.text.x = element_text(angle = 90,
                                    hjust = 1,
                                    vjust = 0.5),
        plot.title = element_text(hjust = 0.5))

```

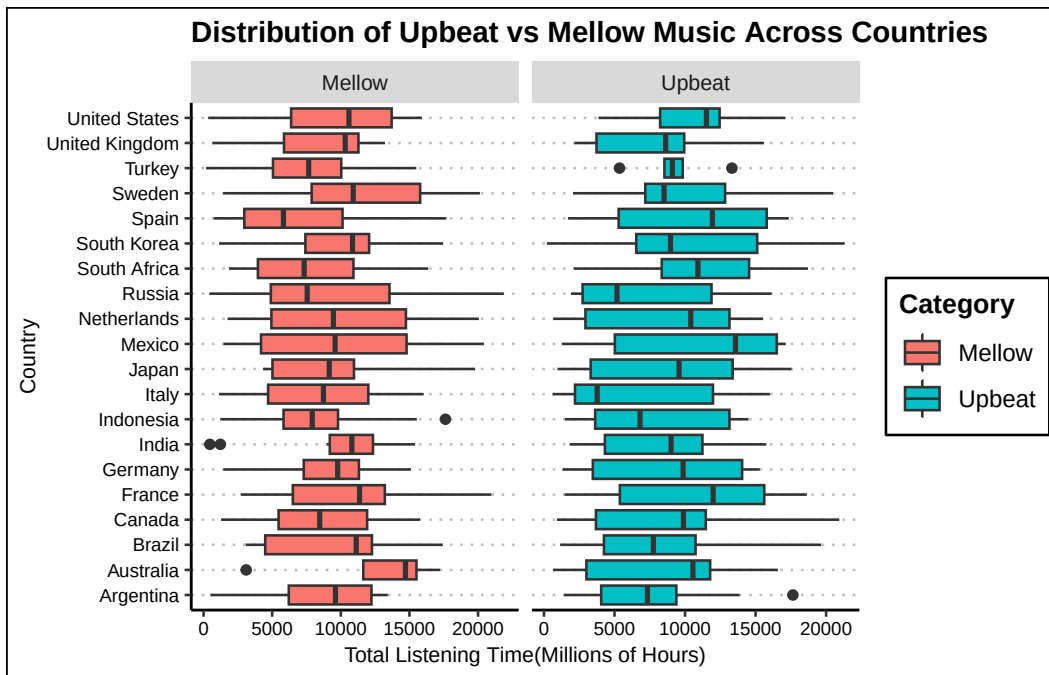


Overall, most countries show declines in GDP, social support, and dystopia scores from 2023 to 2024, indicating worsening economic and social conditions alongside higher perceived hardship.

In contrast, freedom and healthy life expectancy (HLE) generally increased across countries, suggesting improvements in personal autonomy and health. Changes in corruption and generosity are mixed, with no uniform global trend.

3. Distributions for the Upbeat vs Mellow Music Across Countries

```
ggplot(data = spotify_dataset_modified,
       mapping = aes(x = Country,
                     y = Total_hours_Streamed,
                     fill = Category)) +
  geom_boxplot() +
  theme_clean(base_size = 10) +
  labs(title = "Distribution of Upbeat vs Mellow Music Across Countries",
       x = "Country",
       y = "Total Listening Time(Millions of Hours)") +
  coord_flip() +
  facet_wrap(~Category, scales = "free_x")
```



Across nearly all countries, Upbeat tracks have higher median listening times and wider variability than Mellow tracks, indicating stronger and more diverse engagement with upbeat music.

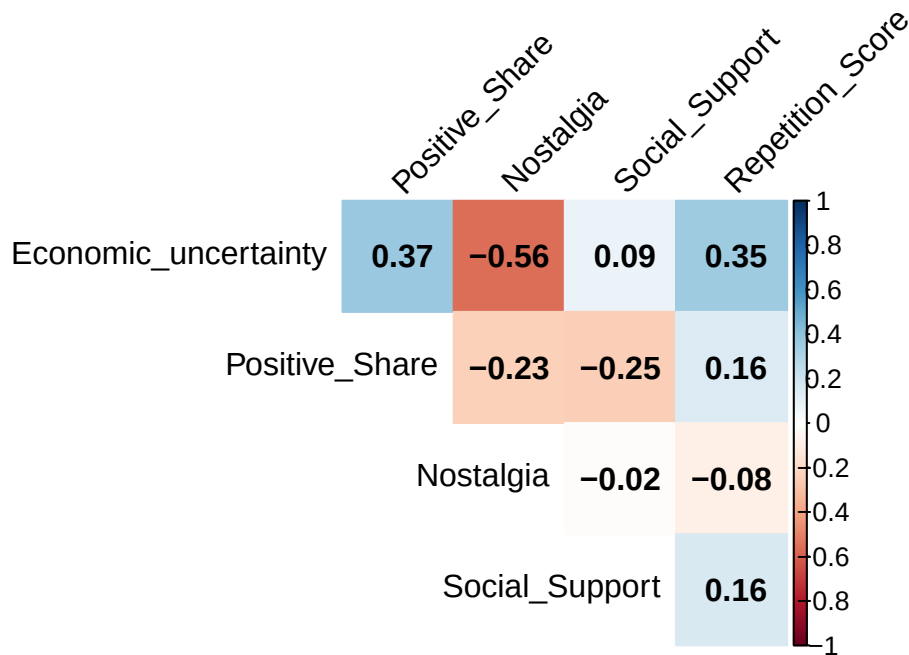
Analysis

In order to analyse the data, first a correlation matrix was constructed using the `corrplot` package. Then the correlations were visualized using scatterplots and the `geom_smooth` function to see the line of best fit.

```
cor_data <- joined_genre_well_being %>%
  select(economics_uncertainty_index,
         social_support_index,
         Repetition_Score,
         Positive_Hours,
         nostalgia_index) %>%
  rename(Positive_Share = Positive_Hours,
         Economic_uncertainty = economics_uncertainty_index,
         Social_Support = social_support_index,
         Nostalgia = nostalgia_index)

cor_matrix <- cor(cor_data, use = "pairwise.complete.obs", method = "pearson")

corrplot(cor_matrix,
         method = "color",
         type = "upper",
         order = "hclust",
         diag = FALSE,
         addCoef.col = "black",
         tl.col = "black",
         tl.srt = 45)
```



As you can see in the correlation matrix: 1. Economic uncertainty and repetition listening were positively correlated which supports H1 that in times of national economic stress, people prefer listening to comfort music rather than exploring. 2. Economic uncertainty and positive listening share has a moderate positive correlation which supports the h2 where a rise in economic uncertainty leads to consumers choosing more positive music as a way to better cope. Social support and the positive listening share also had a weak negative correlation which supported this finding and implies that low social support leads to higher positive music consumption. 3. Economic uncertainty and nostalgia are negatively correlated which does not support H3 where it was hypothesized that in times of economic stress, consumers will choose older more nostalgic music. The study found that it was the complete opposite. This could be due to the fact that the dataset only includes music released between 2018 to 2023 where as nostalgic music may include music from a lot earlier.

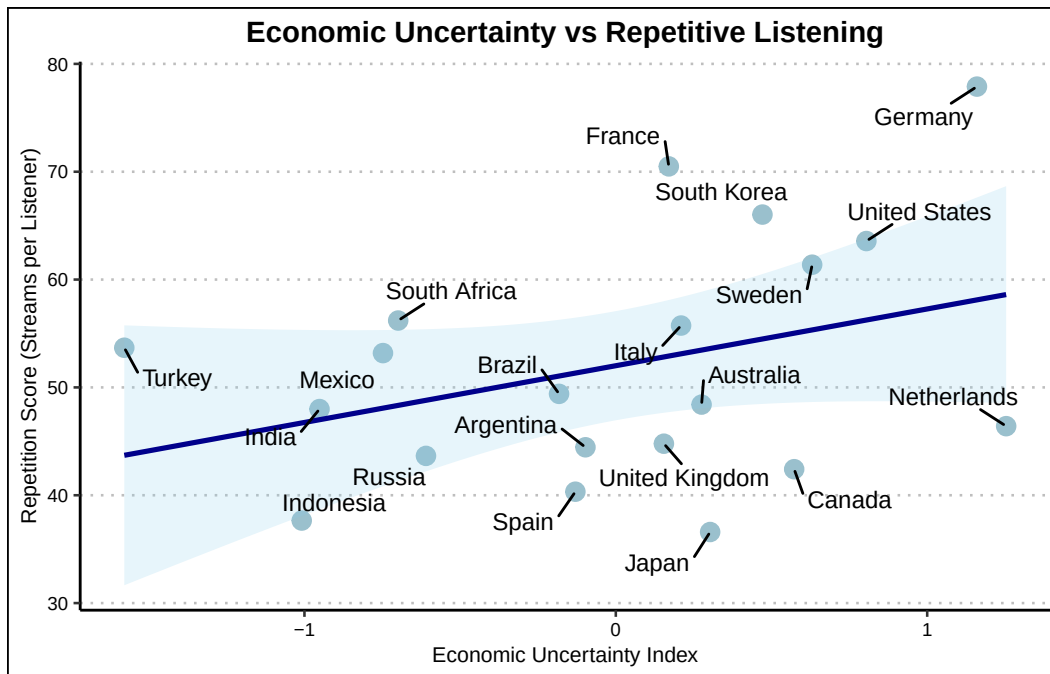
These correlations are visualized below. The labels were added using the ggrepel package.

```
# H1
ggplot(data = joined_genre_well_being,
       mapping = aes(x = economics_uncertainty_index,
                     y = Repetition_Score)) +
  geom_point(color = "lightblue3", size = 3) +
  geom_smooth(method = "lm",
             color = "darkblue",
             fill = "skyblue",
             alpha = 0.2) +
```

```

geom_text_repel(mapping = aes(label = Country),
                size = 3,
                box.padding = 0.5) +
labs(title = "Economic Uncertainty vs Repetitive Listening",
     x = "Economic Uncertainty Index",
     y = "Repetition Score (Streams per Listener)") +
theme_clean(base_size = 10) +
theme(plot.title = element_text(hjust = 0.5))

```

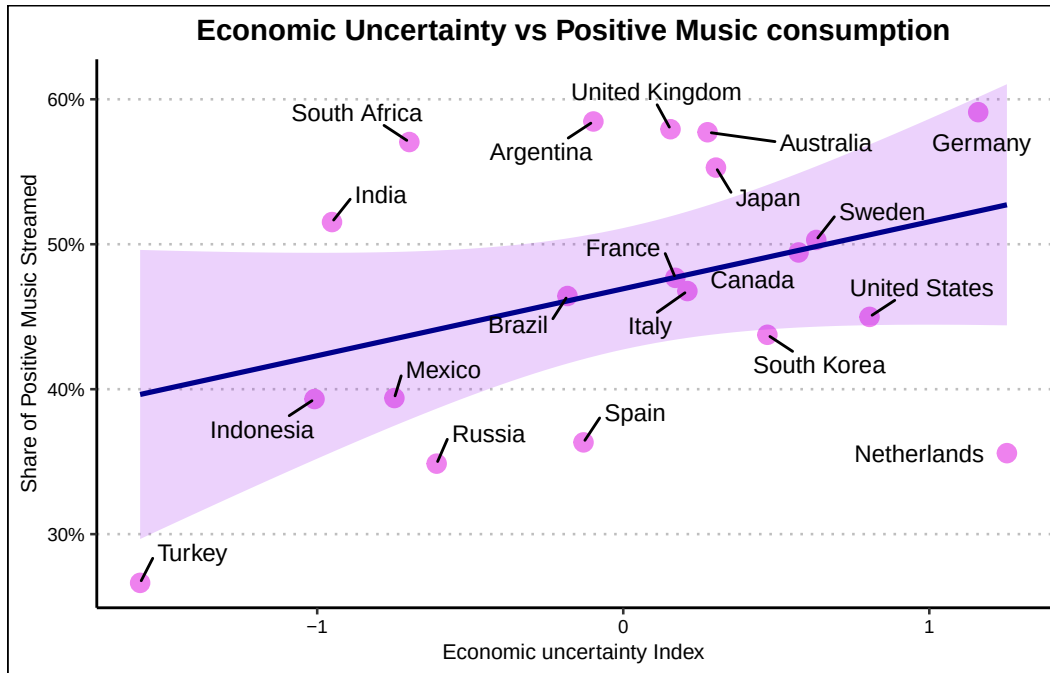


```

# H2
ggplot(data = joined_genre_well_being,
       mapping = aes(x = economics_uncertainty_index,
                     y = Positive_Hours)) +
geom_point(color = "violet", size = 3) +
geom_smooth(method = "lm",
            color = "darkblue",
            fill = "purple",
            alpha = 0.2) +
geom_text_repel(aes(label = Country),
                size = 3,
                box.padding = 0.5) +
scale_y_continuous(labels = scales::percent) +

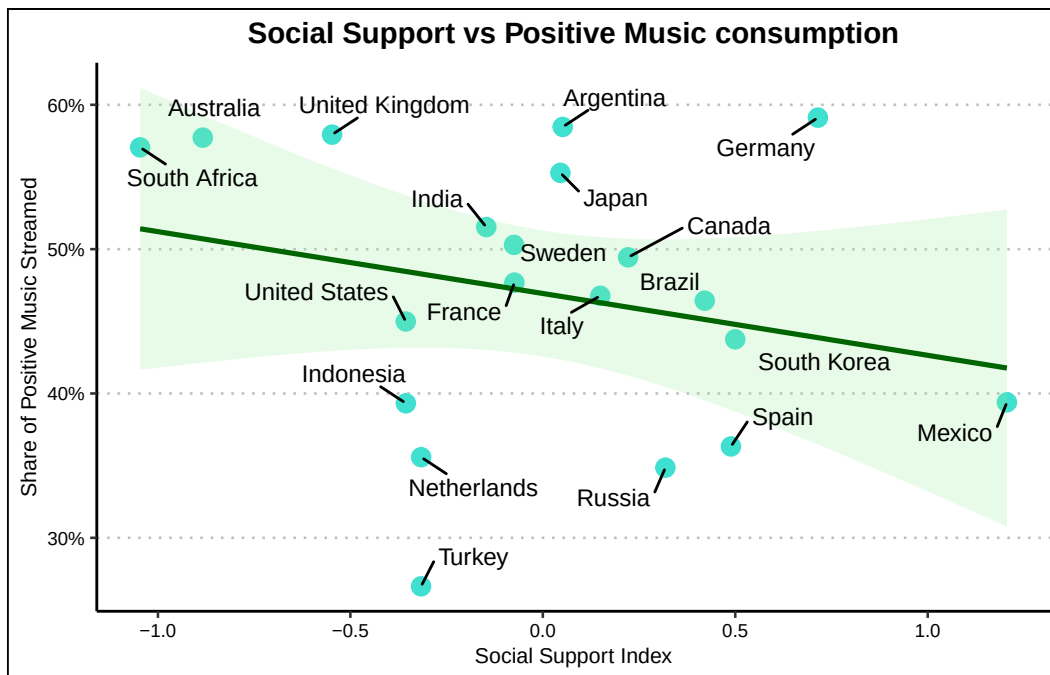
```

```
labs(title = "Economic Uncertainty vs Positive Music consumption",
     x = "Economic uncertainty Index",
     y = "Share of Positive Music Streamed") +
theme_clean(base_size = 10) +
theme(plot.title = element_text(hjust = 0.5))
```

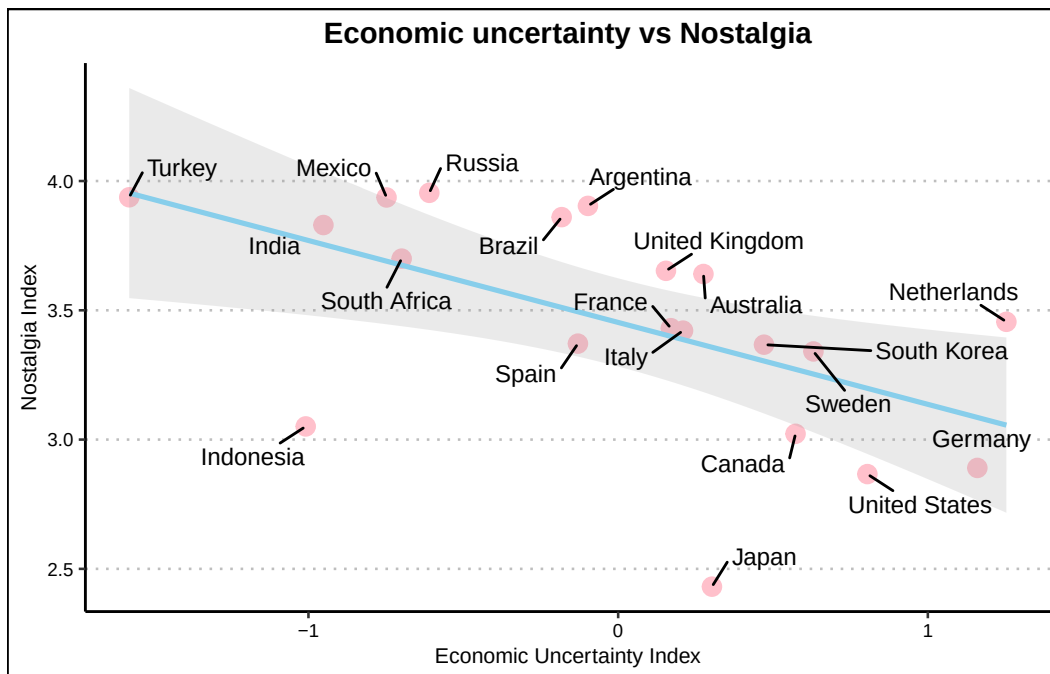


```
ggplot(data = joined_genre_well_being,
       mapping = aes(x = social_support_index,
                     y = Positive_Hours)) +
geom_point(color = "turquoise", size = 3) +
geom_smooth(method = "lm",
            color = "darkgreen",
            fill = "lightgreen",
            alpha = 0.2) +
geom_text_repel(aes(label = Country),
                size = 3,
                box.padding = 0.5) +
scale_y_continuous(labels = scales::percent) +
labs(title = "Social Support vs Positive Music consumption",
     x = "Social Support Index",
     y = "Share of Positive Music Streamed") +
theme_clean(base_size = 10) +
```

```
theme(plot.title = element_text(hjust = 0.5))
```



```
# H3
ggplot(data = joined_genre_well_being,
       mapping = aes(x = economics_uncertainty_index,
                     y = nostalgia_index)) +
  geom_point(color = "pink", size = 3) +
  geom_smooth(method = "lm",
             color = "skyblue",
             alpha = 0.2) +
  geom_text_repel(aes(label = Country),
                 size = 3,
                 box.padding = 0.5) +
  labs(title = "Economic uncertainty vs Nostalgia",
       x = "Economic Uncertainty Index",
       y = "Nostalgia Index") +
  theme_clean(base_size = 10) +
  theme(plot.title = element_text(hjust = 0.5))
```



```
#H1
cor.test(x = cor_data$Economic_uncertainty,
         y = cor_data$Repetition_Score,
         alternative = "greater")
```

Pearson's product-moment correlation

```
data: cor_data$Economic_uncertainty and cor_data$Repetition_Score
t = 1.5958, df = 18, p-value = 0.06397
alternative hypothesis: true correlation is greater than 0
95 percent confidence interval:
 -0.03113072  1.00000000
sample estimates:
      cor
0.3520614
```

```
#H2
cor.test(x = cor_data$Economic_uncertainty,
         y = cor_data$Positive_Share,
         alternative = "greater")
```

Pearson's product-moment correlation

```
data: cor_data$Economic_uncertainty and cor_data$Positive_Share
t = 1.6931, df = 18, p-value = 0.05383
alternative hypothesis: true correlation is greater than 0
95 percent confidence interval:
 -0.009759377  1.000000000
sample estimates:
      cor
0.3706496
```

```
cor.test(x = cor_data$Social_Support,
         y = cor_data$Positive_Share,
         alternative = "less")
```

Pearson's product-moment correlation

```
data: cor_data$Social_Support and cor_data$Positive_Share
t = -1.0771, df = 18, p-value = 0.1478
alternative hypothesis: true correlation is less than 0
95 percent confidence interval:
 -1.0000000  0.1466462
sample estimates:
      cor
-0.2460693
```

```
#H3
cor.test(x = cor_data$Economic_uncertainty,
         y = cor_data$Nostalgia)
```

Pearson's product-moment correlation

```
data: cor_data$Economic_uncertainty and cor_data$Nostalgia
t = -2.8495, df = 18, p-value = 0.01064
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8021585 -0.1527108
sample estimates:
```

cor
-0.5575534

When tested for statistical significance:

H1: The positive correlation between economic uncertainty and repetition is not statistically significant since the p-value is greater than 0.05.

H2: The positive correlation between economic uncertainty and positive listening is approaching statistical significance at p-value 0.05383

H3: The correlation between economic uncertainty and nostalgia is statistically significant since the p-value is lesser than 0.05. However, it was a negative correlation which went against the hypothesis.

Conclusion

To conclude, the study examined how shifts in national well-being relate to global music consumption patterns for the year 2024. While economic uncertainty showed a positive relationship with repetitive listening, this association was not statistically significant, providing only directional but not inferential support for H1. Partial support was found for H2, as higher economic uncertainty was associated with a greater share of positive music consumption at a trend level, and lower social support showed a weak negative association with positive music share, together suggesting that upbeat music may serve as a compensatory emotional resource during socioeconomic stress. Contrary to H3, economic uncertainty was significantly associated with lower nostalgia, indicating a shift toward newer rather than older music during periods of instability. This could be due to how nostalgia was operationalized, as the dataset only included songs released between 2018 and 2023. Overall, these findings suggest that national well-being is meaningfully linked to collective music consumption patterns and that genre-level listening behavior may offer a tentative behavioral marker of socioeconomic conditions.

AI Disclosure

AI was used to construct the correlation matrix. It was used to improve the well-being plots to go over and under the line and to add the country labels in the correlation plots. It was also used to compare different ways of creating the indices from the world happiness report data which are then all tested to see which was the best approach. Other than that it was only used to debug the entire code at the end.

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